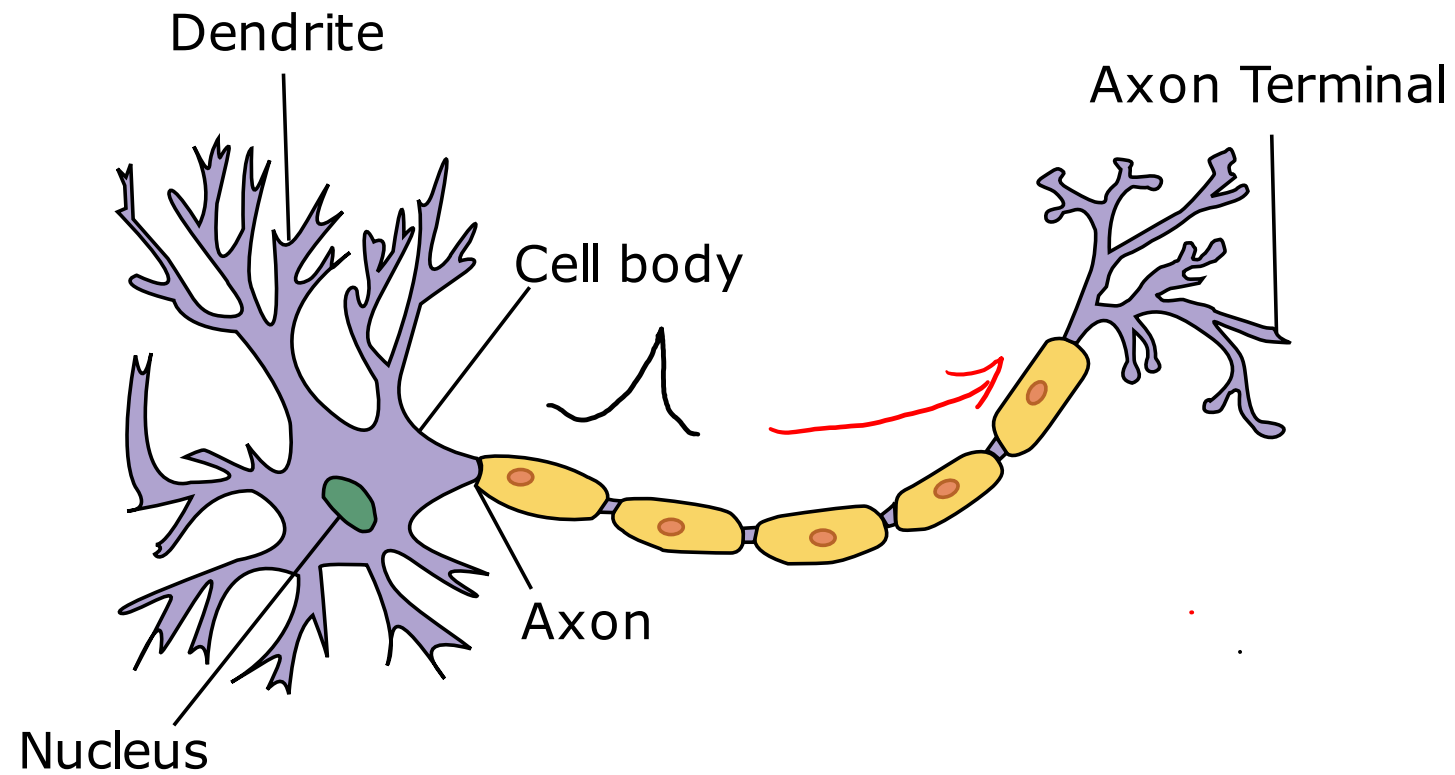


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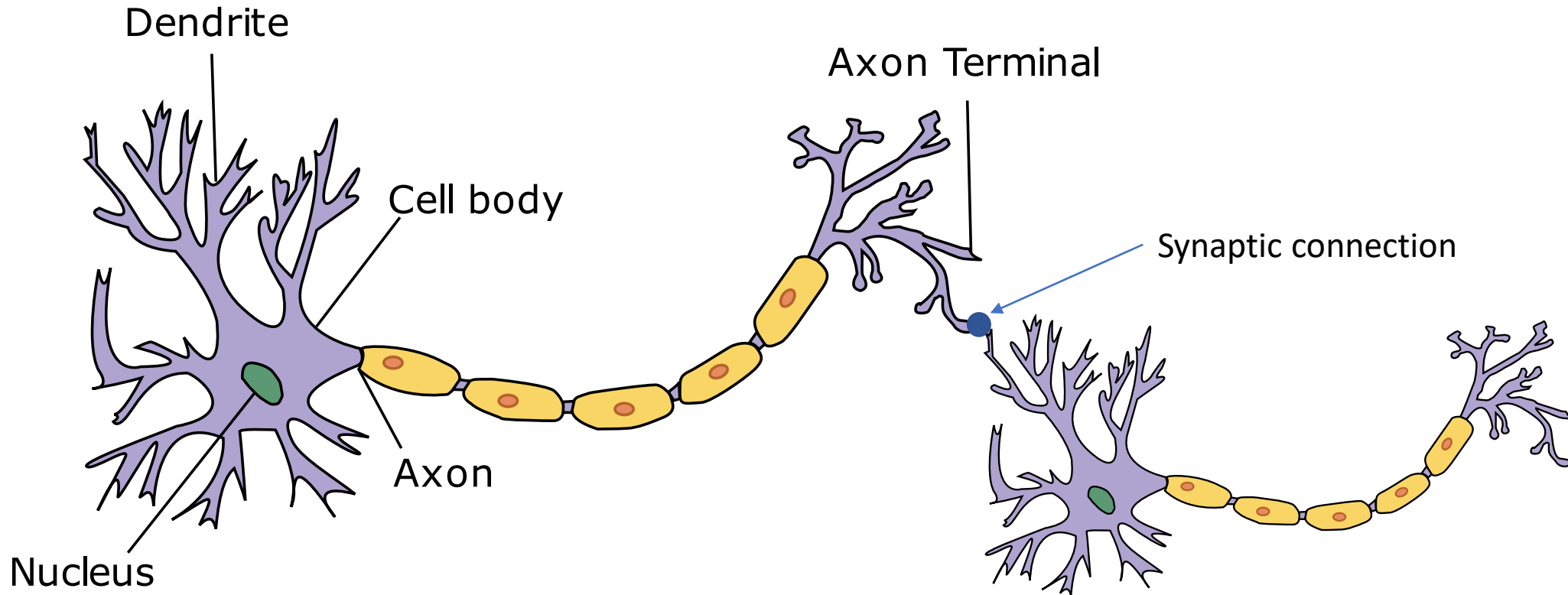
Neural Networks



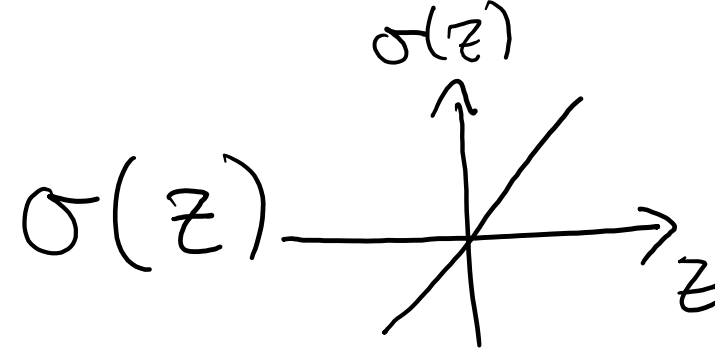
Biological Neurons



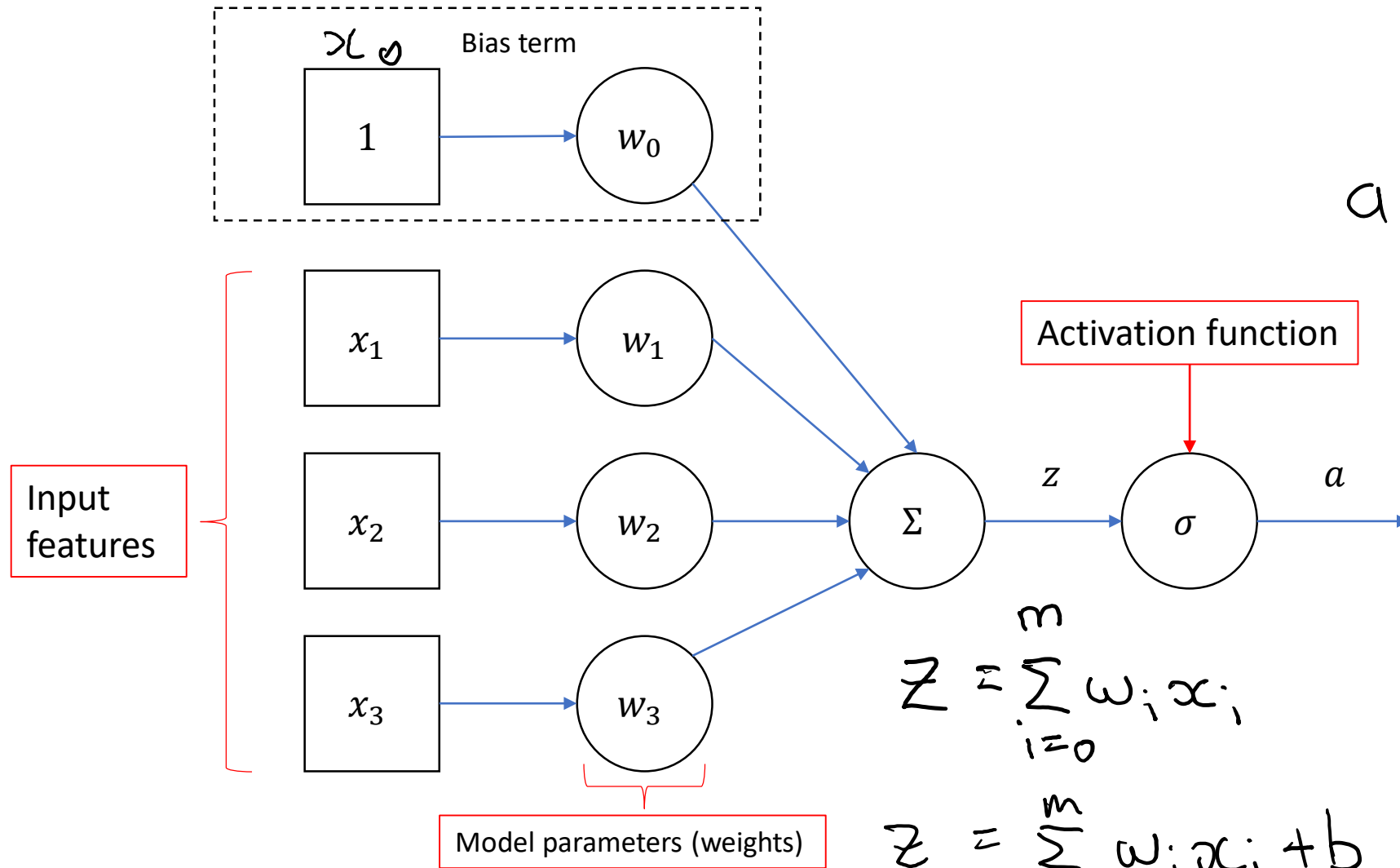
Biological Neural Networks



Artificial Neuron



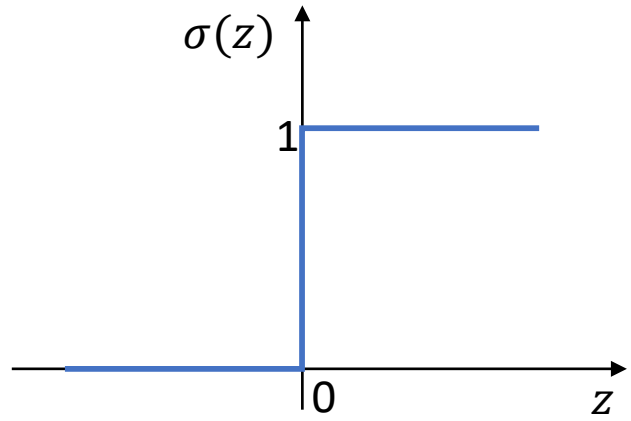
$$a = \sigma(\sum \omega_i x_i + b)$$
$$= \sum \omega_i x_i + b$$



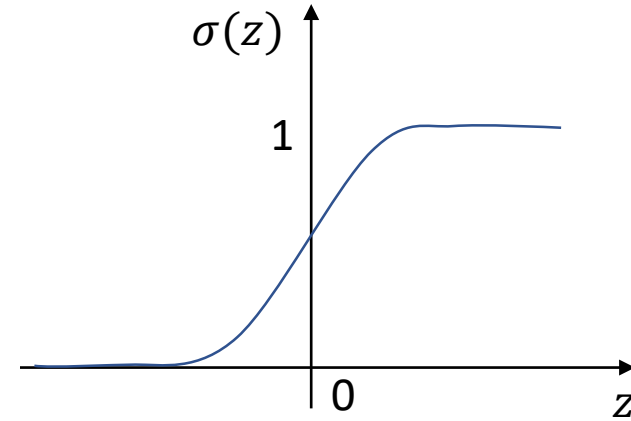
$$z = \sum_{i=0}^m \omega_i x_i$$
$$z = \sum_{i=1}^m \omega_i x_i + b$$



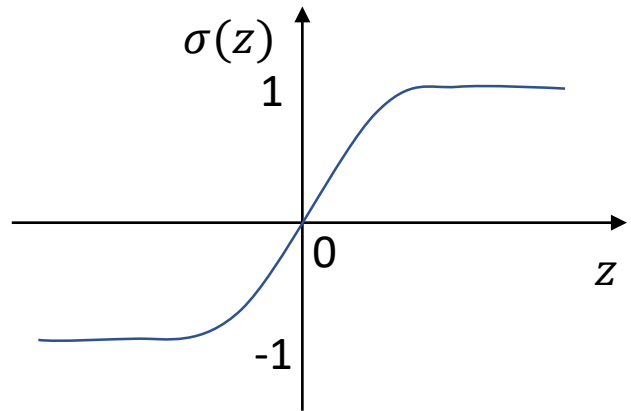
Common Activation functions



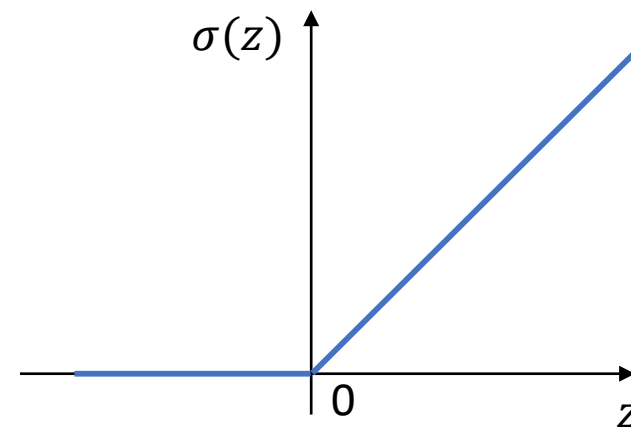
Threshold/Heaviside step



Logistic function: $\sigma(z) = \frac{1}{1+e^{-z}}$



Hyperbolic tangent: $\sigma(z) = \tanh(z)$



Rectified Linear Unit (ReLU): $\sigma(z) = \max(0, z)$

Next up:

Fully-connect, multi-layer, feed forward neural networks

